

The potential for agentic AI usage across various industries

Healthcare: Agentic AI improves diagnosis through dynamic learning from actual cases and autonomous prediction optimisation. It customises treatments through ongoing adaptation of plans in accordance with patient response. In predictive medicine, AI agents actively propose interventions, and in hospital management, they dynamically allocate resources. The technology improves drug discovery via autonomous experimentation and augments telemedicine through smart management of patient interactions and routing critical cases.

Finance: In financial transactions, agentic AI transforms portfolio management by executing real-time trading decisions and automating loan underwriting. It also enhances risk determination and fraud detection by examining huge amounts of data with unprecedented speed and accuracy, cutting review cycles and increasing decision transparency.

Information technology: Agentic AI streamlines IT operations, from proactive infrastructure management to AI-driven service desks. It accelerates software development via text-to-code generation, code review automation, and legacy system modernisation. AI agents liberate human effort, making it more efficient so engineers have more time for challenging problem-solving.

Manufacturing: In production, agentic AI drives intelligent factories, streamlining robot fleet simulations, inventory optimisation, and production scheduling. It optimises logistics with real-time route optimisation and assists with sustainability by reducing energy usage and waste, resulting in quicker production cycles and less expense.

E-commerce: Agentic AI personalises customer experiences via behaviour analysis across digital channels. It supports dynamic

pricing models, optimises marketing campaigns, and enhances customer service with AI-driven virtual agents answering queries in an instant or routing cases astutely.

Agentic AI vs robotic process automation (RPA)

RPA bots are often used to handle repetitive, routine tasks by following a strict set of pre-defined rules. They don't 'think' in the way agentic AI does—they simply execute the same sequence of actions every time, without considering changes in the environment or learning from past results.

In contrast, agentic AI systems are designed to be more dynamic and adaptable. Instead of relying on a fixed script, these agents can sense what's happening around them, make informed decisions, and adjust their strategies as conditions evolve. Imagine having a digital assistant that not only follows your instructions, but can also spot new opportunities, respond to unexpected challenges, and learn from mistakes over time. That's the key difference — unlike RPA, agentic AI is not just about doing a task; it's about understanding the task and finding smarter ways to do it.

Core components of agentic AI architecture

Agentic AI systems are made up of several modules that work together to generate intelligent and adaptive behaviour. Each module deals with a specific task but is related to the others, thus allowing the system to perceive its world, process information, make decisions, and respond in an appropriate manner. Let's look at each of these components in detail for better understanding.

Perception module: This is a sense system of the AI agent that receives and processes information from its surrounding environment. It can acquire data through different sensors -- cameras, microphones, touch sensors, or digital APIs. As an example, an autonomous car uses cameras for road signs, LIDAR for environmental mapping, and microphones to detect some form of voice command.

- **Feature extraction:** Raw sensory information is used to pull out significant features. So, in face recognition, an AI system pulls out features such as the distance between two eyes or the shape of a jawline, which are crucial to recognise a person.

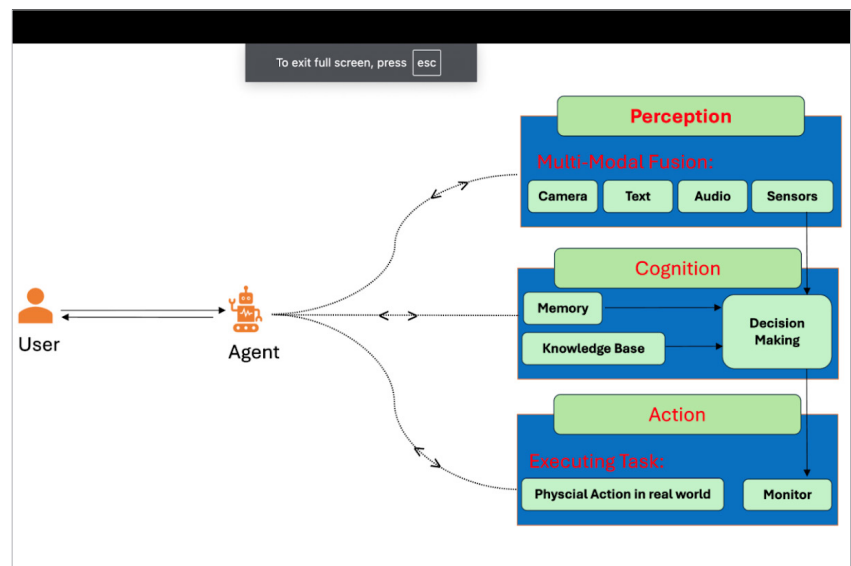


Figure 3: Core components of agentic AI architecture